

Characterizing and Correcting Non-Standard Arithmetic in GPUs

for Scientific Computing

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Motivation

- Modern computing relies on hardware accelerators for high performance
 - GPUs have embraced these specialized units, commonly known as Tensor Cores, to accelerate matrix operations of the form:

$$D = A \times B + C$$

A ₀₀	A ₀₁	A ₀₂	A ₀₃
A ₁₀	A ₁₁	A ₁₂	A ₁₃
A ₂₀	A ₂₁	A ₂₂	A ₂₃
A ₃₀	A ₃₁	A ₃₂	A ₃₃

B ₀₀	B ₀₁	B ₀₂	B ₀₃
B ₁₀	B ₁₁	B ₁₂	B ₁₃
B ₂₀	B ₂₁	B ₂₂	B ₂₃
B ₃₀	B ₃₁	B ₃₂	B ₃₃

C ₀₀	C ₀₁	C ₀₂	C ₀₃
C ₁₀	C ₁₁	C ₁₂	C ₁₃
C ₂₀	C ₂₁	C ₂₂	C ₂₃
C ₃₀	C ₃₁	C ₃₂	C ₃₃

- Until recently, the precise semantics of these units were proprietary and effectively a “black box”

Motivation

- Through recent work by Fasi¹, Li², Mantas³, and Valpey⁴ that we now understand how these hardware function
 - However, sometimes simply understanding the semantics is not enough
- Early work from Fasi and Mantas demonstrated a phenomenon emergent from the non-standard behavior of Tensor Core accumulation: **non-monotonicity**
- Non-monotonicity can violate basic assumptions in numerically sensitive code, affecting convergence, error bounds, and stability

[1] Massimiliano Fasi, Nicholas J Higham, Mantas Mikaitis, and Srikara Pranesh. Numerical Behavior of NVIDIA Tensor Cores. *PerJ Computer Science*, 7:e330, 2021.

[2] Xinyi Li, Ang Li, Bo Fang, Katarzyna Swirydowicz, Ignacio Laguna, and Ganesh Gopalakrishnan. FTTN: FeatureTargeted Testing for Numerical Properties of nvidia & AMD Matrix Accelerators. In 2024 IEEE/ACM 24th International Symposium on Cluster, Cloud and Internet Computing (CCGrid). IEEE, 2024.

[3] Mantas Mikaitis. Monotonicity of multi-term floating-point adders. *IEEE Transactions on Computers*, 73(6):1531–1543, 2024.

[4] Benjamin Valpey, Xinyi Li, Sreepathi Pai, and Ganesh Gopalakrishnan. An SMT Formalization of MixedPrecision Matrix Multiplication: Modeling Three Generations of Tensor Cores. In *NASA Formal Methods Symposium*, pages 360–379, Cham, Jun 2025. Springer Nature Switzerland.

Motivation

Consider an addition of 3 terms:

$$1 + 2 + 8$$

...and a second addition where addends are larger:

$$1 + 2 + 9$$

- **Non-monotonicity** occurs when increasing an operand decreases result's magnitude
- i.e. it is possible for $(1 + 2 + 9) < (1 + 2 + 8)$ to be true

Motivation

- TC results are “good enough” for machine learning
 - TC often not “good enough” for scientific computing

Motivation

- The internal arithmetic of these accelerators evolves with each generation
- Important to formalize the precise conditions for non-monotonicity in tensor cores in an effort to
 - Understand input ranges that may trigger this behavior
 - Inform hardware designs in future generations such that these dangerous emergent phenomenon like are guarded against

Preliminaries

- Floating point values consist of a sign (s), exponent (e), and an m-bit mantissa. The term $1.m$ is known as the significand
 - Commonly expressed as $(-1)^s \cdot 1.m \cdot 2^e$
- **ulp** (unit in the last place) – the value of the mantisa's least significant bit

sign		exp		impl		mant
0		20		1		01101

← ulp

Preliminaries

- Tensor cores generally vary generation by generation in two ways:
 1. n : number of terms added in one block
 2. p : the number of extra ‘padding’ bits appended to the end of the mantissa

Architecture	Addends (n)	Padding (p)
Volta	5	0
Turing	5	1
Ampere	9	1
Hopper	17	2

Preliminaries

- A tensor core performs its operations in 4 primary steps:
 - $A[i,k] \cdot B[k,j]$ are computed exactly and held in FP32
 - Because exact, can simplify $A[i,k] \cdot B[k,j]$ terms to be $T_1 \dots T_i$
 - A max exponent is determined from all $T_1 \dots T_i$ and the C term and all terms are right-shifted to align to this exponent
 - The aligned significands are summed at once in an extended significand with p extra bits of padding precision
 - Results are normalized and bits beyond mantissa are truncated and lost

Preliminaries ($n = 4$, $p = 1$)

sign		exp		impl		mant		padding		lost
0		20		1		01101		0		
0		18		1		01110		0		
0		16		1		11100		0		
0		17		1		01111		0		

Preliminaries (n = 4, p = 1)

sign		exp		impl		mant		padding		lost
0		20		1		01101		0		
0		18		1		01110		0		
0		16		1		11100		0		
0		17		1		01111		0		

Preliminaries ($n = 4$, $p = 1$)

sign		exp		impl		mant		padding		lost
0		20		1		01101		0		
0		20		0		01011		1		0
0		20		0		00011		1		100
0		20		0		00101		1		11

Preliminaries ($n = 4$, $p = 1$)

sign	exp	impl	mant	padding	lost
0	20	1	01101	0	
0	20	0	01011	1	0
0	20	0	00011	1	100
0	20	0	00101	1	11

Preliminaries ($n = 4$, $p = 1$)

sign	exp	impl	mant	padding	lost
0	20	1	01101	0	
0	20	0	01011	1	
0	20	0	00011	1	
0	20	0	00101	1	

Preliminaries (n = 4, p = 1)

sign		exp		impl		mant		padding		lost
------	--	-----	--	------	--	------	--	---------	--	------

0		20		1		01101		0		
---	--	----	--	---	--	-------	--	---	--	--

0		20		0		01011		1		
---	--	----	--	---	--	-------	--	---	--	--

0		20		0		00011		1		
---	--	----	--	---	--	-------	--	---	--	--

0		20		0		00101		1		
---	--	----	--	---	--	-------	--	---	--	--

SUM	-----									
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Preliminaries (n = 4, p = 1)

sign	exp	impl	mant	padding	lost
------	-----	------	------	---------	------

0	20	1	01101	0	
---	----	---	-------	---	--

0	20	0	01011	1	
---	----	---	-------	---	--

0	20	0	00011	1	
---	----	---	-------	---	--

0	20	0	00101	1	
---	----	---	-------	---	--

SUM -----

0	20	10	00001	1	
---	----	----	-------	---	--

Preliminaries (n = 4, p = 1)

sign		exp		impl		mant		padding		lost
------	--	-----	--	------	--	------	--	---------	--	------

0		20		1		01101		0		
---	--	----	--	---	--	-------	--	---	--	--

0		20		0		01011		1		
---	--	----	--	---	--	-------	--	---	--	--

0		20		0		00011		1		
---	--	----	--	---	--	-------	--	---	--	--

0		20		0		00101		1		
---	--	----	--	---	--	-------	--	---	--	--

SUM -----

0		21		1		00000		1		1
---	--	----	--	---	--	-------	--	---	--	---

Preliminaries (n = 4, p = 1)

sign		exp		impl		mant		padding		lost
------	--	-----	--	------	--	------	--	---------	--	------

0		20		1		01101		0		
---	--	----	--	---	--	-------	--	---	--	--

0		20		0		01011		1		
---	--	----	--	---	--	-------	--	---	--	--

0		20		0		00011		1		
---	--	----	--	---	--	-------	--	---	--	--

0		20		0		00101		1		
---	--	----	--	---	--	-------	--	---	--	--

SUM -----

0		21		1		00000		1		1
---	--	----	--	---	--	-------	--	---	--	---

Preliminaries (n = 4, p = 1)

sign	exp	impl	mant	padding	lost
------	-----	------	------	---------	------

0	20	1	01101	0	
---	----	---	-------	---	--

0	20	0	01011	1	
---	----	---	-------	---	--

0	20	0	00011	1	
---	----	---	-------	---	--

0	20	0	00101	1	
---	----	---	-------	---	--

SUM -----

0	21	1	00000		
---	----	---	-------	--	--

Non-monotonicity

- Now that we know how these units work, let's try to investigate a case of non-monotonicity...

Non-Monotonicity ($n = 4$, $p = 0$)

sign	exp	impl	mant	lost
0	24	1	00000	0
0	18	1	00000	0
0	18	1	00000	0
0	18	1	00000	0

Non-Monotonicity (n = 4, p = 0)

sign	exp	impl	mant	lost
0	24	1	00000	
0	18	1	00000	
0	18	1	00000	
0	18	1	00000	

Non-Monotonicity (n = 4, p = 0)

sign	exp	impl	mant	lost
0	24	1	00000	0
0	24	0	00000	1
0	24	0	00000	1
0	24	0	00000	1

Non-Monotonicity (n = 4, p = 0)

sign	exp	impl	mant	lost
0	24	1	00000	0
0	24	0	00000	1
0	24	0	00000	1
0	24	0	00000	1

Non-Monotonicity (n = 4, p = 0)

sign		exp		impl		mant		lost
------	--	-----	--	------	--	------	--	------

0		24		1		00000		
---	--	----	--	---	--	-------	--	--

0		24		0		00000		
---	--	----	--	---	--	-------	--	--

0		24		0		00000		
---	--	----	--	---	--	-------	--	--

0		24		0		00000		
---	--	----	--	---	--	-------	--	--

SUM	-----							
-----	-------	--	--	--	--	--	--	--

Non-Monotonicity (n = 4, p = 0)

sign		exp		impl		mant		lost
------	--	-----	--	------	--	------	--	------

0		24		1		00000		
---	--	----	--	---	--	-------	--	--

0		24		0		00000		
---	--	----	--	---	--	-------	--	--

0		24		0		00000		
---	--	----	--	---	--	-------	--	--

0		24		0		00000		
---	--	----	--	---	--	-------	--	--

SUM	-----							
-----	-------	--	--	--	--	--	--	--

0		24		1		00000		
---	--	----	--	---	--	-------	--	--

Non-Monotonicity (n = 4, p = 0)

sign		exp		impl		mant		lost
0		23		1		11111		0
0		18		1		00000		0
0		18		1		00000		0
0		18		1		00000		0

Non-Monotonicity ($n = 4$, $p = 0$)

sign		exp		impl		mant		lost
0		23		1		11111		0
0		18		1		00000		0
0		18		1		00000		0
0		18		1		00000		0

Non-Monotonicity (n = 4, p = 0)

sign		exp		impl		mant		lost
0		23		1		11111		0
0		23		0		00001		0
0		23		0		00001		0
0		23		0		00001		0

Non-Monotonicity (n = 4, p = 0)

sign		exp		impl		mant		lost
------	--	-----	--	------	--	------	--	------

0		23		1		11111		0
---	--	----	--	---	--	-------	--	---

0		23		0		00001		0
---	--	----	--	---	--	-------	--	---

0		23		0		00001		0
---	--	----	--	---	--	-------	--	---

0		23		0		00001		0
---	--	----	--	---	--	-------	--	---

SUM	-----							
-----	-------	--	--	--	--	--	--	--

Non-Monotonicity (n = 4, p = 0)

sign		exp		impl		mant		lost
------	--	-----	--	------	--	------	--	------

0		23		1		11111		0
---	--	----	--	---	--	-------	--	---

0		23		0		00001		0
---	--	----	--	---	--	-------	--	---

0		23		0		00001		0
---	--	----	--	---	--	-------	--	---

0		23		0		00001		0
---	--	----	--	---	--	-------	--	---

SUM	-----							
-----	-------	--	--	--	--	--	--	--

0		23		10		00010		0
---	--	----	--	----	--	-------	--	---

Non-Monotonicity (n = 4, p = 0)

sign		exp		impl		mant		lost
------	--	-----	--	------	--	------	--	------

0		23		1		11111		0
---	--	----	--	---	--	-------	--	---

0		23		0		00001		0
---	--	----	--	---	--	-------	--	---

0		23		0		00001		0
---	--	----	--	---	--	-------	--	---

0		23		0		00001		0
---	--	----	--	---	--	-------	--	---

SUM -----

0		24		1		00001		0
---	--	----	--	---	--	-------	--	---

Non-Monotonicity

- Non-monotonicity described by literature to be resultant from a class of tensor core-like adders computing n-term sums without the normalization of intermediate quantities

Non-Monotonicity

- Non-monotonicity described by literature to be resultant from a class of tensor core-like adders computing n-term sums without the normalization of intermediate quantities
- Is there a space within this class that is guaranteed to be safe from non-monotonicity?
 - Lack of intermediate normalization is much of what makes TC efficient
 - → Can we keep this general schema while still correcting for its problematic behavior?

Non-Monotonicity

- INTUITION: We can see from the example when given a large term and $n - 1$ small terms
 - When the large term's exponent decreases, the smaller terms re-enter the representable range of the large term and into the accumulator

Non-Monotonicity

- Non-monotonic behavior occurs in a Tensor Core accumulator if the sum of the smaller terms that “flow back” into the accumulator is greater than the decrease in the large term, without loss of generality: $T_1 - T_1'$

Non-Monotonicity

- Non-monotonic behavior occurs in a Tensor Core accumulator if the sum of the smaller terms that “flow back” into the accumulator is greater than the decrease in the large term, without loss of generality: $T1 - T1'$
- INTUITION: A Tensor Core accumulator is $m + p$ bits wide. These p bits are appended onto the lower end of the mantissa. This means for smaller terms to “flow in and out” of the large term, they must fall below the padding bits too

Lets try!

Non-Monotonicity (n = 4, p = 1)

sign	exp	impl	mant	padding	lost
0	24	1	00000	0	
0	18	1	00000	0	
0	18	1	00000	0	
0	18	1	00000	0	

Non-Monotonicity (n = 4, p = 1)

sign	exp	impl	mant	padding	lost
0	24	1	00000	0	
0	24	0	00000	1	
0	24	0	00000	1	
0	24	0	00000	1	

Non-Monotonicity (n = 4, p = 1)

sign	exp	impl	mant	padding	lost
0	24	1	00000	0	
0	24	0	00000	1	
0	24	0	00000	1	
0	24	0	00000	1	

SUM -----

Non-Monotonicity (n = 4, p = 1)

sign		exp		impl		mant		padding		lost
------	--	-----	--	------	--	------	--	---------	--	------

0		24		1		00000		0		
---	--	----	--	---	--	-------	--	---	--	--

0		24		0		00000		1		
---	--	----	--	---	--	-------	--	---	--	--

0		24		0		00000		1		
---	--	----	--	---	--	-------	--	---	--	--

0		24		0		00000		1		
---	--	----	--	---	--	-------	--	---	--	--

SUM -----

0		24		1		00001		1		
---	--	----	--	---	--	-------	--	---	--	--

Non-Monotonicity (n = 4, p = 1)

sign	exp	impl	mant	padding	lost
------	-----	------	------	---------	------

0	24	1	00000	0	
---	----	---	-------	---	--

0	24	0	00000	1	
---	----	---	-------	---	--

0	24	0	00000	1	
---	----	---	-------	---	--

0	24	0	00000	1	
---	----	---	-------	---	--

SUM -----

0	24	1	00001	1	
---	----	---	-------	---	--

Non-Monotonicity (n = 4, p = 1)

sign		exp		impl		mant		padding		lost
------	--	-----	--	------	--	------	--	---------	--	------

0		24		1		00000		0		
---	--	----	--	---	--	-------	--	---	--	--

0		24		0		00000		1		
---	--	----	--	---	--	-------	--	---	--	--

0		24		0		00000		1		
---	--	----	--	---	--	-------	--	---	--	--

0		24		0		00000		1		
---	--	----	--	---	--	-------	--	---	--	--

SUM -----

0		24		1		00001				
---	--	----	--	---	--	-------	--	--	--	--

Non-Monotonicity ($n = 4$, $p = 1$)

sign	exp	impl	mant	padding	lost
0	23	1	11111	0	
0	18	1	00000	0	
0	18	1	00000	0	
0	18	1	00000	0	

Non-Monotonicity (n = 4, p = 1)

sign	exp	impl	mant	padding	lost
0	23	1	11111	0	
0	18	1	00000	0	
0	18	1	00000	0	
0	18	1	00000	0	

Non-Monotonicity (n = 4, p = 1)

sign	exp	impl	mant	padding	lost
0	23	1	11111	0	
0	23	0	00001	0	
0	23	0	00001	0	
0	23	0	00001	0	

Non-Monotonicity (n = 4, p = 1)

sign	exp	impl	mant	padding	lost
0	23	1	11111	0	
0	23	0	00001	0	
0	23	0	00001	0	
0	23	0	00001	0	

SUM -----

Non-Monotonicity (n = 4, p = 1)

sign		exp		impl		mant		padding		lost
------	--	-----	--	------	--	------	--	---------	--	------

0		23		1		11111		0		
---	--	----	--	---	--	-------	--	---	--	--

0		23		0		00001		0		
---	--	----	--	---	--	-------	--	---	--	--

0		23		0		00001		0		
---	--	----	--	---	--	-------	--	---	--	--

0		23		0		00001		0		
---	--	----	--	---	--	-------	--	---	--	--

SUM -----

0		23		10		00010		0		
---	--	----	--	----	--	-------	--	---	--	--

Non-Monotonicity (n = 4, p = 1)

sign	exp	impl	mant	padding	lost
------	-----	------	------	---------	------

0	23	1	11111	0	
---	----	---	-------	---	--

0	23	0	00001	0	
---	----	---	-------	---	--

0	23	0	00001	0	
---	----	---	-------	---	--

0	23	0	00001	0	
---	----	---	-------	---	--

SUM -----

0	23	1	00001	0	
---	----	---	-------	---	--

Non-Monotonicity (n = 4, p = 1)

sign	exp	impl	mant	padding	lost
------	-----	------	------	---------	------

0	23	1	11111	0	
---	----	---	-------	---	--

0	23	0	00001	0	
---	----	---	-------	---	--

0	23	0	00001	0	
---	----	---	-------	---	--

0	23	0	00001	0	
---	----	---	-------	---	--

SUM -----

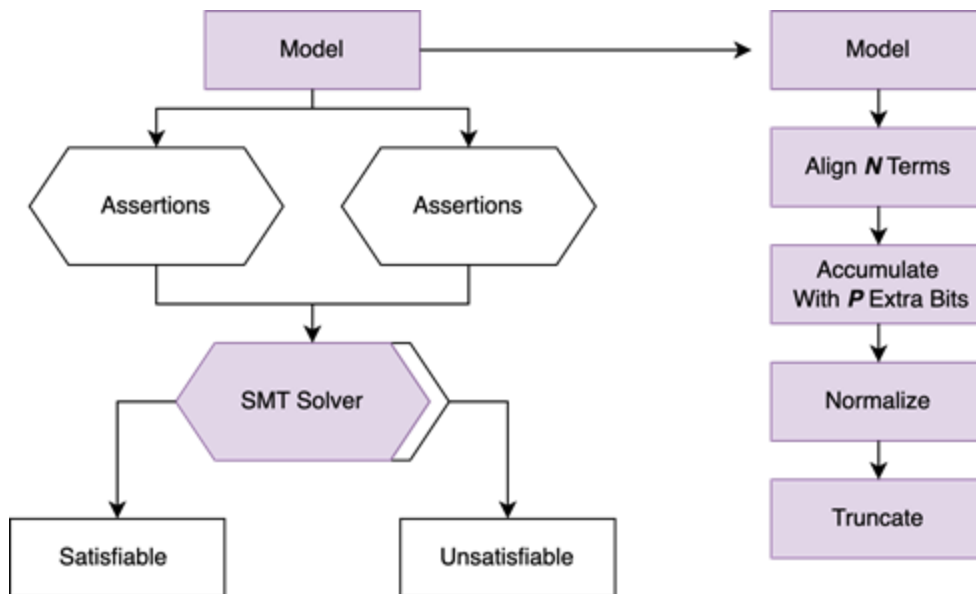
0	23	1	00001		
---	----	---	-------	--	--

Non-Monotonicity

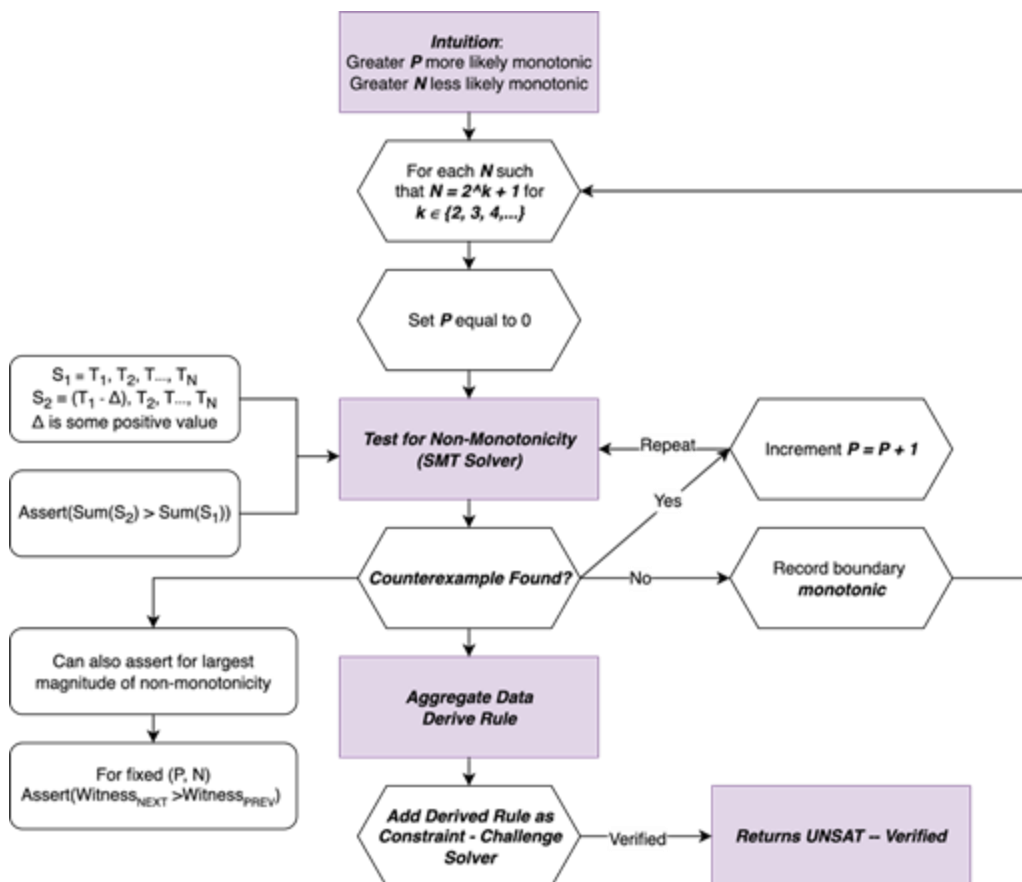
- No longer non-monotonic for this one example
 - But this is just for one example, is there a way to know for all cases?

SMT

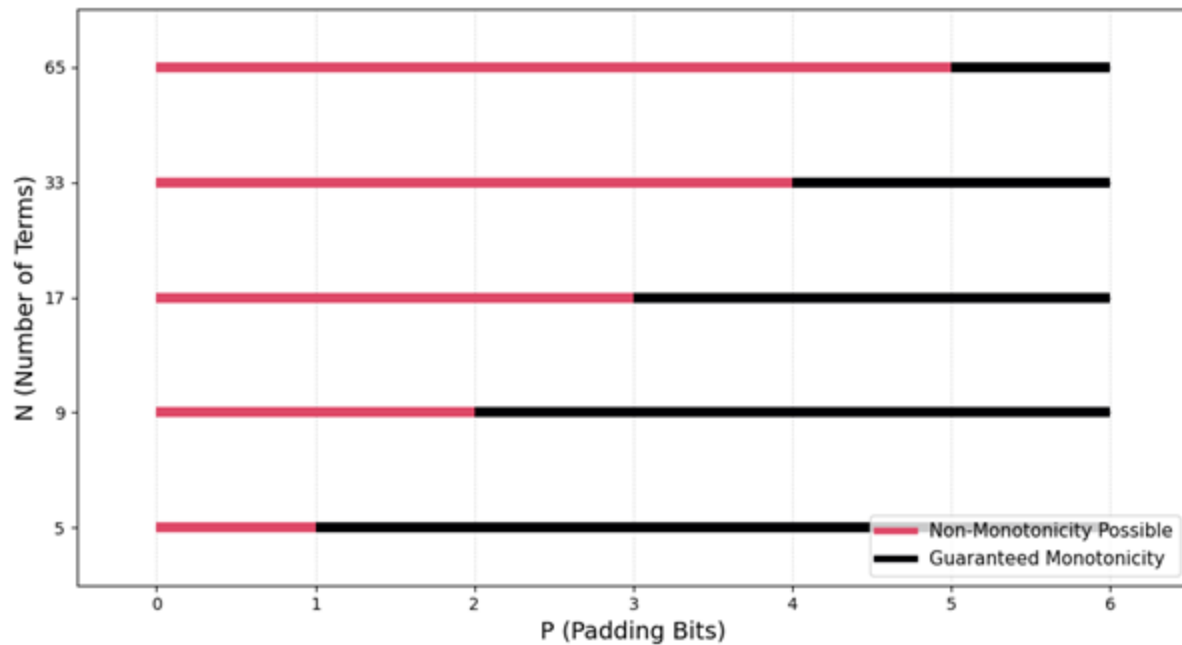
- SMT Solvers can be powerful in modeling arithmetic schemes
- We can encode each operation that the tensor core performs as constraints on bit-vectors



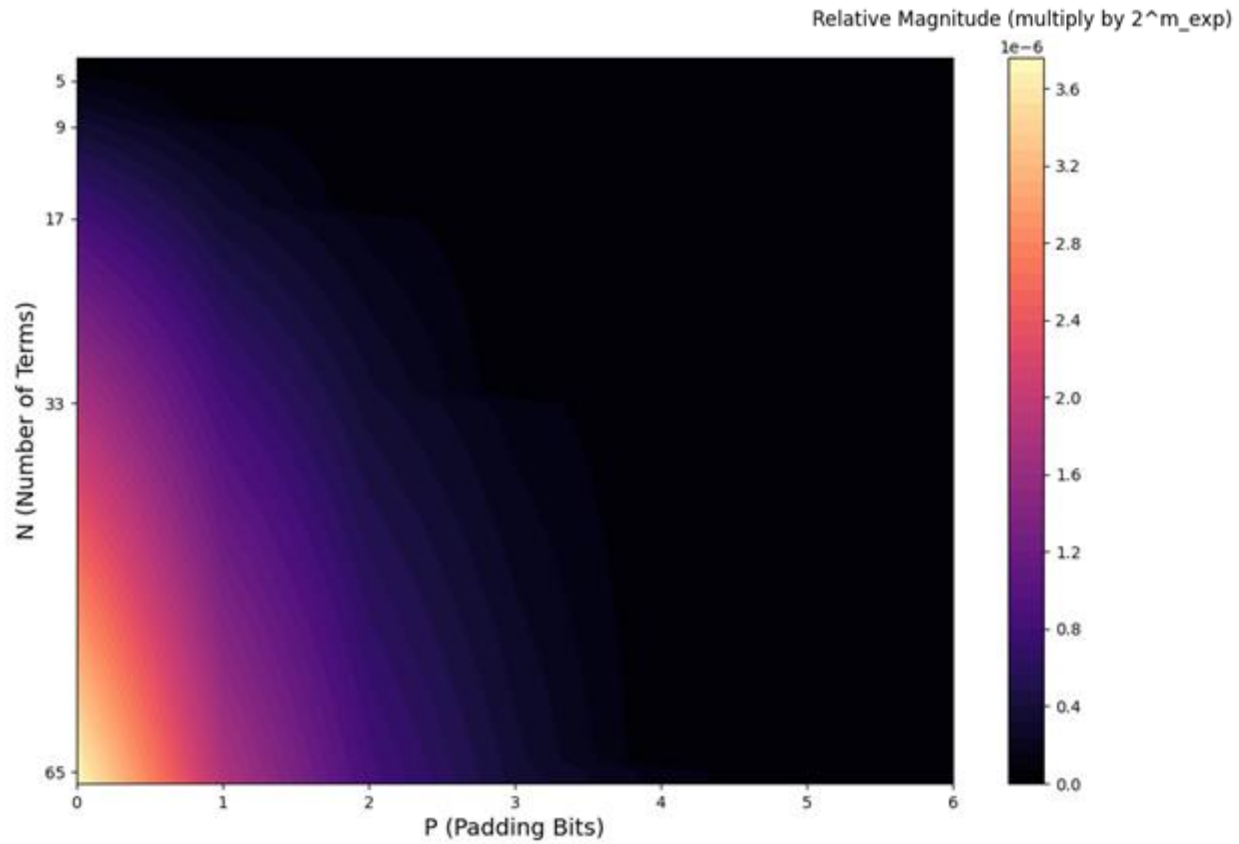
Process



Results



Results



Results

Architecture	Addends (n)	Padding (p)	Result
Volta	5	0	Susceptible (True)
Turing	5	1	Monotonic (False)
Ampere	9	1	Susceptible (True)
Hopper	17	2	Susceptible (True)

Results

Architecture	Addends (n)	Padding (p)	Result
Volta	5		Susceptible (True)
Turing			Monotonic (False)
Amper			Susceptible (True)
Hopper		2	Susceptible (True)

$$p \geq \lfloor \log(n - 1) - 2 \rfloor$$

Takeaways

- YES! There is a space of hardware configurations that fall within this class of Tensor Core-like adders that also does not allow for non-monotonicity
 - The existing Turing architecture falls under this classification
- We have only discussed non-monotonicity here
 - What about the larger bounds of general error that come inherent to TC
 - Can we correct for these too?
 - → Come talk with me, we can discuss :)

Thank You



